

Element I

The objective of this testing procedure was to evaluate the design through a series of performance and durability tests. The idea was focused around removability, ease of use, and ergonomics which were all previously provided by the client as some current issues with the design.

For our testing process, we utilized the clamps used on the prototype and tested the maximum amount of tension required to damage the prototype. Although this may be a somewhat unconventional type of testing, we were able to test to get a glimpse of the strength of the toggle clamps against a cardboard design. As of now, this was the best way to test this design because we don't have a final design for the box to test it with, nor do we have a very solidified prototype for the box. Surprisingly, the box could withstand the force of about 200 N, which means that it could withstand much more if properly secured. Another thing that we learned while testing was that when testing the strength, we ended up pulling the toggle clamps out of the box, instead of the toggle clamps popping off or loosening.

In terms of testing the final product, the procedure was very brief and included a trial of testing and revising. Corbin Wolfgang, a member of another group, worked on the door and delivered it to us on April 14, 2025. Since the addition of both a handle on the end and a new guillotine door, we have been working to perfect the functionality of the doors. The current issue revolves around the fact that one of the doors doesn't slide all the way down because 1. The door is somewhat warped and 2. The gap that the door slides into is extremely narrow and forces the user to jam the door into the slot on the bottom.

Below, you can see the design working in



The biggest modification that may come from this repeated testing of the guillotine door by pulling it up and down repeatedly would be the possibly reducing the friction.

While the testing plan addresses key elements of the design, there are inherent limitations due to the constraints of using prototypes and the evolving nature of the design process. One primary limitation is the use of cardboard in the prototype testing. Although it serves as a reasonable substitute for initial testing, cardboard does not accurately reflect the final material properties, such as strength, durability, and resistance to wear. Therefore, while the tests on the cardboard prototype can provide some insight into the design's performance, they may not fully represent how the system will behave when constructed with the intended materials, such as metal or reinforced plastics.

Another limitation is the inability to conduct long-term durability testing at this stage. Since the design is still in its prototyping phase, repeated and prolonged use of the system, especially under conditions simulating months of handling by zookeepers, cannot be fully assessed. The potential wear and tear on the box connection system, the guillotine door mechanism, and the toggle clamps will need to be tested further after the final materials and design specifications are finalized. This is especially important for

assessing the friction behavior of the guillotine door, which may change with long-term use, potentially affecting its ease of operation.

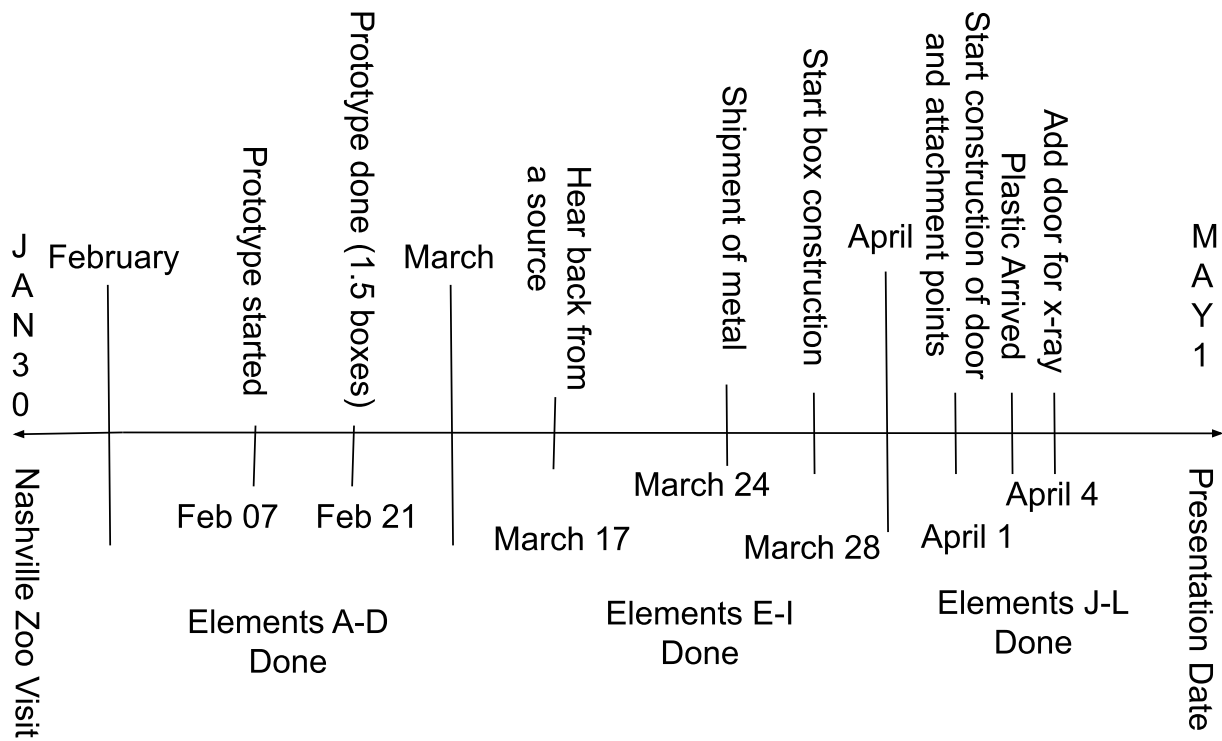
Moreover, while the current tests on the system's strength, usability, and ease of use are crucial, they are based on controlled scenarios. Real-world testing, such as field trials with zookeepers actively interacting with the system, could reveal unforeseen issues related to ergonomics, operational efficiency, or unforeseen design flaws. This could include issues with the system's adaptability, where the connection points might become misaligned with repeated use, or if the design is found to be cumbersome during routine tasks. These factors may not be evident in the initial testing and will require iterative adjustments based on feedback from real-world use.

The testing and data collection conducted thus far have provided important insights into the overall functionality and performance of the Komodo dragon containment system. The guillotine door mechanism, box connection system, and toggle clamps have been tested under various conditions, with results indicating promising strength and usability. The ability of the box to withstand 200 N of force is particularly noteworthy, demonstrating the system's potential for durability when finalized with the intended materials. The friction levels of the guillotine door and the potential modifications to improve its sliding mechanism will be key factors to address before finalizing the design.

The data collected from these initial tests will guide further refinement and modifications to the design. For example, reducing friction in the guillotine door will enhance its ease of use, while adjustments to the toggle clamps and box connection system may improve long-term reliability and ease of reconfiguration. As the project moves forward, continued testing and feedback will be essential to ensure the system meets the necessary performance standards, particularly in terms of safety, usability, and durability.

In conclusion, the prototype testing and data collection process has provided valuable insights that will guide the refinement of the Komodo dragon containment system. Although some limitations exist due to the use of prototypes and the need for long-term testing, the results thus far demonstrate a solid

foundation for the design. With further testing, particularly in real-world settings, the design can be adjusted to meet all required specifications and ultimately provide a functional and reliable solution for the containment of Komodo dragons.



Considering the timeline, the testing will definitely have to be perfect and quick because of the time constraints. Here we have the timeline for the expected progress.